Shail Jadav

Indian Institute of Technology Gandhinagar
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https://shailjadav.github.io/

Education

PhD Candidate in Mechanical Engineering July-2018 to Dec-2023
Indian Institute of Technology Gandhinagar
Thesis: "*Human-Inspired Learning Controllers and Motion Planners for Robotic Manipulators*"
Adviser: Prof. Harish PM

Bachelors in Biomedical Engineering July 2013- April 2017
Gujarat Technological University
Second rank in college and third rank in university

Professional Experience

Visiting Research Scholar May 2023 -October 2023
Technical University of Vienna (TU Wien)
Project: *Shared Autonomy for Human Robot Interaction*
Adviser: Prof. Dongheui Lee & Prof. Christian Ott

Visiting Research Scholar May 2019 -July 2019
The University of Texas at Austin
Project: *Development of ankle cuff for gait trainer robot*
Adviser: Prof. James Sulzer

Project associate October 2017 -June 2018
Indian Institute of Technology Gandhinagar
Developed a device [PDEYE] to detect early onset of Parkinson's disease based on pupillary light reflex& conducted clinical study for the device
Adviser: Prof. Harish PM & Dr. Vruntang Kumar

Project assistant July 2017-October 2017
Indian Institute of Technology Gandhinagar
Designed a device [PDEYE] to detect early onset of Parkinson's disease based on pupillary light reflex
Adviser: Prof. Harish PM & Dr. Vruntang Kumar

Summer Intern June 2017-July 2017
Indian Institute of Technology Gandhinagar
Developed a remote control high-frequency vibration stimulation belt for rest tremor suppression in Parkinson's disease
Adviser: Prof. Harish PM & Dr. Vruntang Kumar

Biomedical Engineer April 2017-June 2017
AIMS Hospital, Ahmedabad
Leader in the technical support during surgeries and implementation of the new medical equipment.
Quality assurance and quality control of medical devices

Teaching Experience

Graduate Teaching Fellow [Mechatronics] August 2022 – December 2023
Indian Institute of Technology Gandhinagar
Taught undergraduate course on mechatronics along with Prof. Madhu Vadali

Graduate Teaching Fellow [Control Theory] January 2022 – April 2022
Indian Institute of Technology Gandhinagar
Taught undergraduate course on control theory along with Prof. Madhu Vadali

Teaching Assistant
Indian Institute of Technology Gandhinagar
ME LAB II: Enabled students to do "learning-by-doing"
Writing: Enabled students for scientific writing

Grants

Overseas Research Fellowship (~INR 650000) May 2023 – October 2023
Indian Institute of Technology Gandhinagar
Shared Autonomy for Human Robot Interaction
Outcome: ICRA 24 paper

NIDHI PRAYAS grant for product development (~INR 700000) July 2022 - December 2023
Department of Science and Technology (Govt. Of India)
Development of analog adaptive motor driver for robots
Outcome: Motor drivers in commercialization phase

Student Travel Grant | SPARC (~INR 400000) May 2019 – July 2019
Department of Science and Technology (Govt. Of India)
Study of Locomotor Adaptation Using a Single degree of freedom Gait Trainer
Outcome: Journal paper

IEEE RAS Student Travel Grant (~ 250000$) May 2024
IEEE Robotics & automation Society
To attend the best robotics conference ICRA in Japan

IITGN Travel Grant (~ INR 100000) May 2024
Indian Institute of Technology Gandhinagar
To ICRA 24 in Japan

Publications

[J1]	Shail Jadav and Harish PM	Configuration and Force-field Aware Variable Impedance Control with Faster Re-learning	Journal of Intelligent & Robotic Systems (Q1) Link
[J2]	Shail Jadav and Harish PM	Utilization of Manipulator Redundancy for Torque Reduction During Force Interaction	ASME Letters in Dynamic Systems and Control Link
[J3]	Shail Jadav, Karthik Karvaje, Sujay Kadam, Vineet Vashista, James Sulzer, Ashish Deshpande, and Harish PM	Kinematic Analysis and Validation of a Customizable Single Degree-of-freedom Gait Trainer Device	ASME Journal of Medical Devices Link
[J4]	Shah Vrutang, Shail Jadav, Sachin Goyal, and Harish PM	A Machine-Learning-Based Method to Detect Degradation of Motor Control Stability with Implications to Diagnosis of Presymptomatic Parkinson's Disease: A Simulation Study	MDPI Applied Sciences Link
[C1]	Shail Jadav, Johannes Heidersberger, Christian Ott, and Dongheui Lee	Shared Autonomy via Variable Impedance Control and Virtual Potential Field for Encoding Human Demonstrations	IEEE International Conference on Robotics and Automation 2024 Link
[C2]	Suyash Patidar, Shail Jadav, and Harish PM	Redundancy in Planar Robotic Manipulator: A Comparison of Redundancy Configurations for Force Production Tasks	IEEE Indian Control Conference 23 Link
[C3]	Shail Jadav, Shubhankar Riswadkar, Sujay Kadam, and Harish PM	Variable Impedance Learning Control with Faster Re-learning and Reduced Initial Errors in Re-perturbation for Robots Operating in Divergent Force Fields	ACM Advances In Robotics 2023 Link
[C4]	Shubhankar Riswadkar, Shail Jadav, and Harish PM	A Novel Approach for Combining Feedback and Feedforward Control in DC Motor Control: A Smooth Switching Strategy for Time-Varying Systems with Noisy Feedback	ACM Advances In Robotics 2023 Link

Reviewer Service

IEEE Transaction on Robotics
IEEE International Conference on Robotics and Automation (ICRA)
ACM Advances in Robotics
IEEE International Conference on Rehabilitation Robotics

Invited talks

[T1] March 2023	Advances in motion control and entrepreneurship	NIT Sikkim
[T2] March 2023	Advances in motion control	PDEU

Awards

[A1] 2023 Winner Regional Finale of Boeing BUILD

[A2] 2017 Runner-up at Google India Hackathon

Relevant Course Work

Human Robot Interaction
Introduction to Robotics
Modern Control Theory
Control Theory
Digital Control Systems
Nonlinear Control
Fundamentals of Neuroscience
Classics in Brain Science

Hobbies

Tinkering with Electronics & Embedded System | Musical Instruments | Cooking | Relaxing in Nature